

## 2026 - Advances SQL : NULL Functions Exercise

1) SELECT employee\_id, name,  
IFNULL(Salary, 0) AS salary-with-default  
FROM employees;

| employee_id | name    | Salary-with-default |
|-------------|---------|---------------------|
| 1           | Alice   | 5000                |
| 2           | Bob     | 0                   |
| 3           | Charlie | 7000                |
| 4           | Dana    | 0                   |

2) SELECT employee\_id, name,  
IFNULL(department, 'Not Assigned') AS department\_name  
FROM Employees;

3) ~~SELECT order\_id, customer\_id~~  
~~FROM Orders~~  
~~WHERE customer\_id IS NULL;~~

2)   
Answer

| employee_id | name    | department_name |
|-------------|---------|-----------------|
| 1           | Alice   | HR              |
| 2           | Bob     | IT              |
| 3           | Charlie | Not Assigned    |
| 4           | Dana    | Finance         |



3) SELECT order\_id, customer\_id  
FROM Orders  
WHERE Customer\_id IS NULL;

| order_id | Customer_id |
|----------|-------------|
| 103      | NULL        |

4) SELECT order\_id, customer\_id, ~~delivery~~  
IFNULL(delivery-status, 'Pending') AS delivery-status  
FROM Orders;

| order_id | Customer_id | delivery-status |
|----------|-------------|-----------------|
| 101      | 201         | 2024-12-01      |
| 102      | 202         | Pending         |
| 103      | NULL        | 2024-12-03      |

5) SELECT student\_id, name,  
IFNULL(grade, 0) AS final\_grade  
FROM Students;

| student_id | name   | Final_grade |
|------------|--------|-------------|
| 1          | Ethan  | 85          |
| 2          | Maya   | 0           |
| 3          | Olivia | 90          |



6) SELECT Count(Student\_Id) AS not-graded-count  
FROM Student  
WHERE grade ISNULL;

| Not-graded-count |
|------------------|
| 1                |

7) SELECT <sup>id,</sup> name, price,  
IFNULL(discount, 0) AS Final-price  
FROM Products;

| Product_Id | name     | Price | Final Price |
|------------|----------|-------|-------------|
| 501        | Keyboard | 25    | 0           |
| 502        | Mouse    | 15    | 5           |
| 503        | Monitor  | 100   | 0           |

8) SELECT Count(customer\_Id) AS missing-email-count  
FROM Customers  
WHERE email ISNULL;

| missing-email-count |
|---------------------|
| 2                   |

9) SELECT customer\_Id, name,  
IFNULL(email, 'No Email') AS email-display  
FROM Customers;



9. ---

| Customer-Id | Name   | email-display   |
|-------------|--------|-----------------|
| 1           | Linda  | No email        |
| 2           | Joseph | joseph@mail.com |
| 3           | Nia    | No email        |

b) SELECT payment\_id,  
IFNULL(method, 'Unknown') AS method\_display,  
IFNULL(status, 'Unknown') AS status  
FROM payments;

| Payment_id | method_display | Status  |
|------------|----------------|---------|
| 301        | Credit         | Unknown |
| 302        | Paypal         | Success |
| 303        | Unknown        | Failed  |

ii) SELECT item\_id, item\_name,  
IFNULL(quantity, 0) AS quantity\_checked  
FROM inventory;

| Item_id | Item-name | Quantity-checked |
|---------|-----------|------------------|
| 1       | Pen       | 0                |
| 2       | Notebook  | 150              |
| 3       | Erasor    | 0                |



12) SELECT emp\_id,  
COALESCE(bonus, commission, 0) AS first-available-reward  
FROM Employee-extra;

| emp_id | first-available-reward |
|--------|------------------------|
| 1      | 300                    |
| 2      | 100                    |
| 3      | 0                      |

13) SELECT ~~class\_id~~, Count(class\_id) AS no-room-count  
FROM Classes  
WHERE room IS NULL;

| No-room-count |
|---------------|
| 2             |

14) SELECT student\_id, date,  
IFNULL(status, 'Not Marked') AS attendance-status  
FROM Attendance;

| Student-id | date       | attendance-status |
|------------|------------|-------------------|
| 1          | 2025-04-01 | Not Marked        |
| 2          | 2025-04-01 | Present           |
| 3          | 2025-04-01 | Absent            |



15) SELECT account\_id,  
 IFNULL(account\_type, 'Unknown') AS type\_display,  
 IFNULL(balance, 0) AS balance\_checked  
 FROM Bank\_accounts;

| <del>Account_id</del> | Account_type | Balance |
|-----------------------|--------------|---------|
| A1                    | Savings      | 0       |
| A2                    | Current      | 5000    |
| A3                    | Unknown      | 2000    |

16) SELECT project\_id, title,  
 IFNULL(start\_date, 'TBD') AS start\_display  
 FROM Projects;

| project_id | title          | Start_date_display |
|------------|----------------|--------------------|
| 1          | Website Revamp | 2025-01-10         |
| 2          | Mobile App     | TBD                |
| 3          | Data Migration | TBD                |

17) SELECT review\_id, product\_id,  
 IFNULL(comment, 'No Comment') AS comment\_display,  
 IFNULL(rating, 0) AS rating\_display  
 FROM Reviews

| review_id | Product_id | Comment_display | rating_display |
|-----------|------------|-----------------|----------------|
| 1         | 501        | Great Product   | 4              |
| 2         | 502        | No Comment      | 0              |
| 3         | 503        | Works fine      | 3              |



18) SELECT supplier\_id, name,  
COALESCE(phone, alt-phone, 'No Contact') AS contact\_number  
FROM Suppliers;

| supplier_id | name          | Contact_number |
|-------------|---------------|----------------|
| 1           | Global Goods  | 123456789      |
| 2           | Best Supplies | 987654321      |
| 3           | ValueScore    | No Contact     |

19) SELECT ~~theme~~ User\_id,  
IFNULL(theme, 'Light') AS theme\_set,  
IFNULL(Language, 'English') AS language\_set,  
IFNULL(timezone, 'UTC') AS timezone\_set  
FROM User\_Settings;

| user_id | theme_set | language_set | timezone_set |
|---------|-----------|--------------|--------------|
| 1       | Light     | English      | UTC          |
| 2       | Dark      | English      | UTC+1        |
| 3       | Light     | English      | UTC          |

20) SELECT record\_id, machine\_id,  
IFNULL(issue, 'Unknown Issue') AS issue\_log,  
IFNULL(technician, 'Not Assigned') AS technician\_name  
FROM Maintenance;

| record_id | machine_id | issue_log     | technician_name |
|-----------|------------|---------------|-----------------|
| 1         | M101       | Overheating   | Not Assigned    |
| 2         | M102       | Unknown Issue | Not Assigned    |
| 3         | M103       | Jammed        | Alex            |